

Student Performance Analysis

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1. Introduction

Education performance is influenced by multiple demographic, social, and behavioural factors. This project analyses student performance data to understand how variables such as gender, parental background, test preparation, sports activity, and study hours impact students' **Math, Reading, and Writing scores**.

2. Objective of the Project

- To analyse the effect of different student-related factors on academic performance.
- To identify patterns and trends in Math, Reading, and Writing scores.
- To compare overall performance between male and female students.
- To derive meaningful insights using data analysis and visualization.

3. Dataset Description

The dataset contains student-level academic records with the following key columns:

- Gender
- EthnicGroup
- ParentEduc
- LunchType
- TestPrep
- ParentMarital
- PracticeSport

- FirstChild
- NrSiblings
- TransportMeans
- WeeklyStudyHours
- MathScore
- ReadingScore
- WritingScore

4. Tools & Libraries Used

- Python
- Pandas-Data handling and cleaning
- NumPy – Numerical operations
- Matplotlib & Seaborn – Data visualization

5. Data Preprocessing

Step 1: Import Required Libraries

- Loaded essential Python libraries for data manipulation and visualization.

Step 2: Read Dataset

- Imported the CSV file into a pandas DataFrame.

Step 3: Handle Missing Values

- Checked for null values in the dataset.
- Removed rows containing missing values to maintain data quality.

Step 4: Feature Engineering

- Created an **AverageScore** column by combining Math, Reading, and Writing scores for overall performance comparison.

6. Exploratory Data Analysis (EDA)

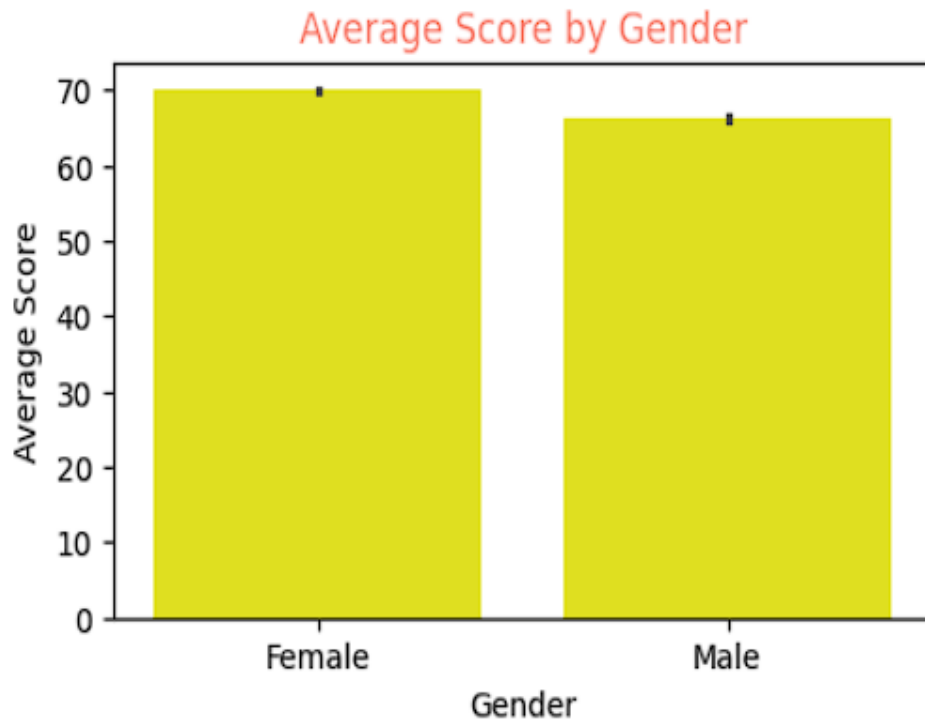
- Analysed score distributions using histograms and box plots
- Identified trends and variations in performance across different categories
- Used bar charts to compare average scores across groups

7. Impact Analysis of Factors on Student Scores (EDA)

➤ Gender

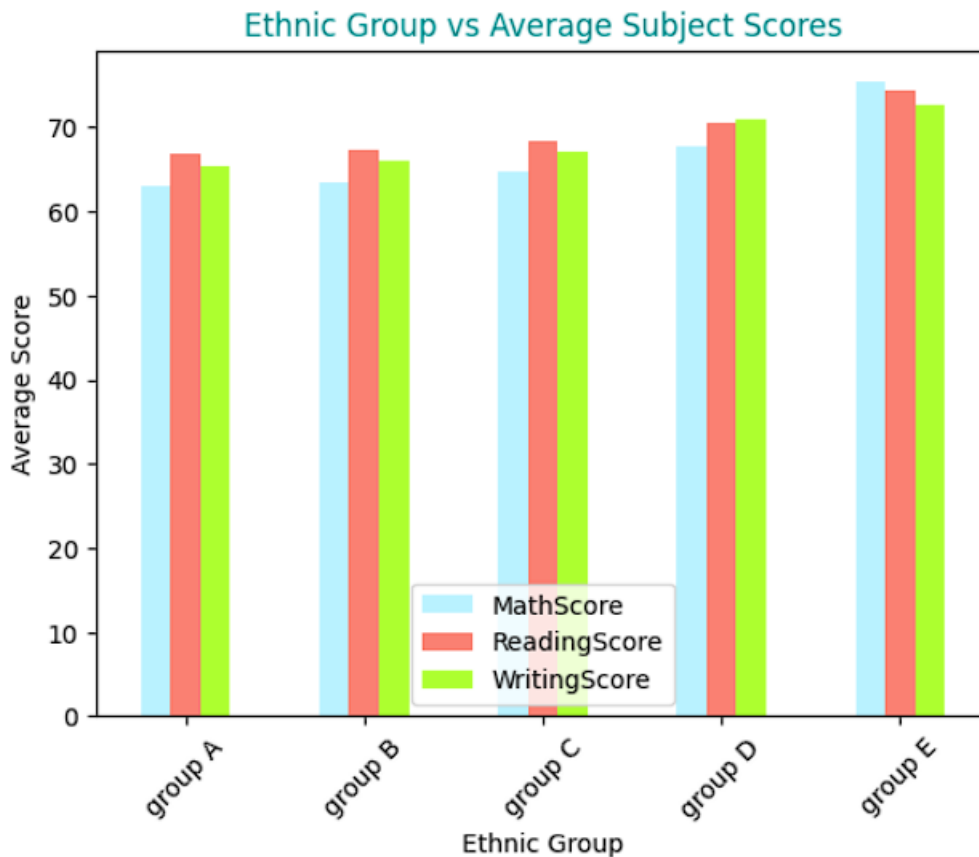
- Compared average Math, Reading, and Writing scores between male and female students.
- Identified which gender performs better overall.
- A bar chart was used to visualize the average score by gender.
- Female students show a higher average score compared to male students.
- This indicates that female students perform better overall across all three subjects.

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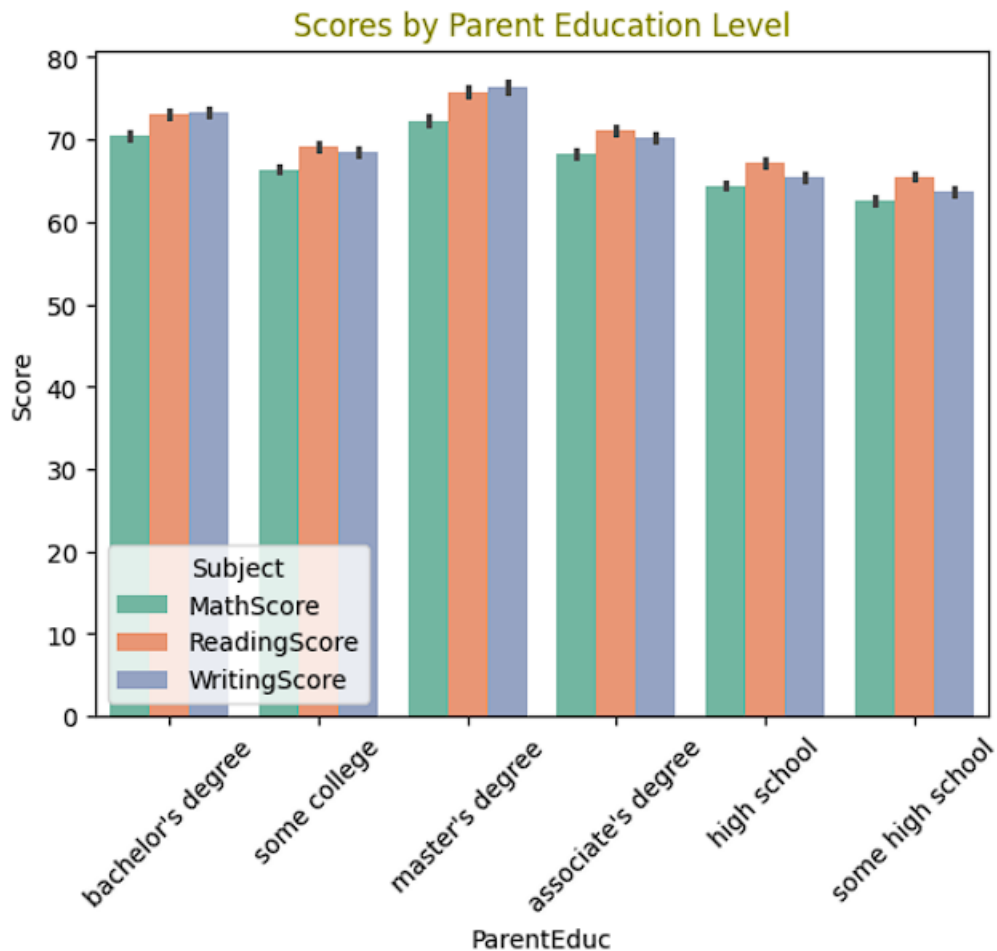
➤ Ethnic Group

- EDA was performed to analyse academic performance across different ethnic groups.
- Average Math, Reading, and Writing scores were compared for each ethnic group.
- Clear variation in subject-wise scores was observed among the ethnic groups.
- Ethnic Group E showed higher overall performance compared to other groups.



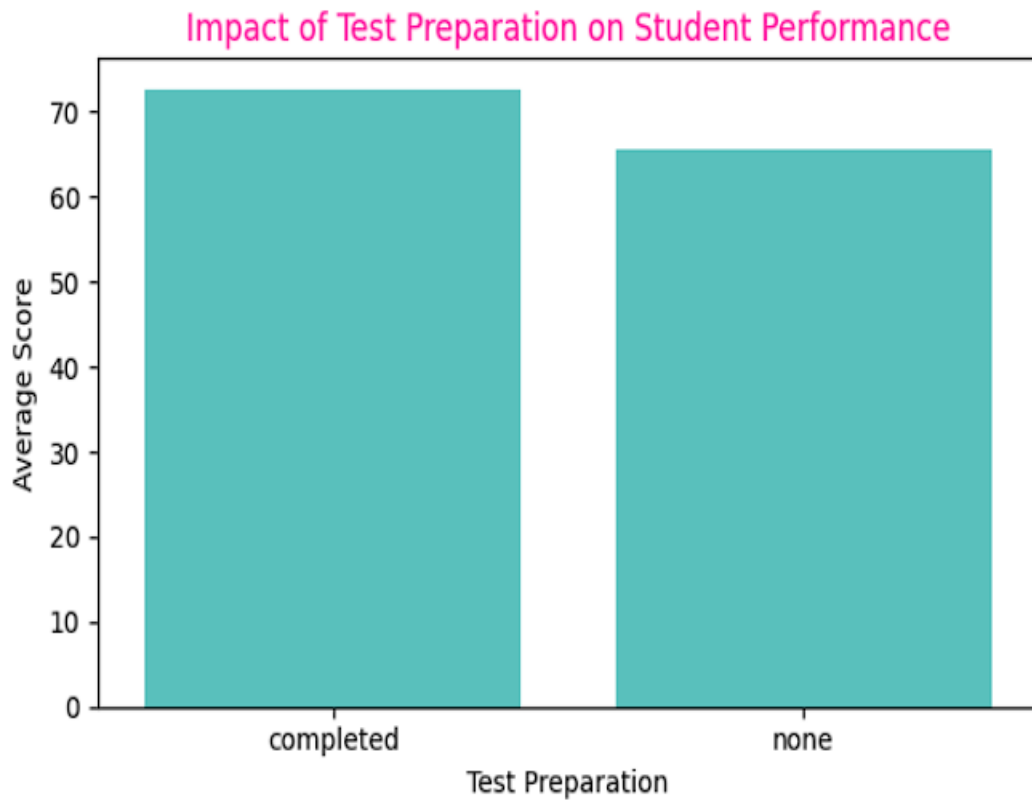
➤ Parent Education

- EDA was conducted to analyse student performance based on parents' education levels.
- Average Math, Reading, and Writing scores were compared across different parental education categories.
- Students whose parents have **higher education levels (bachelor's or master's degree)** show better academic performance.
- The analysis indicates a positive correlation between parental education level and student academic scores.



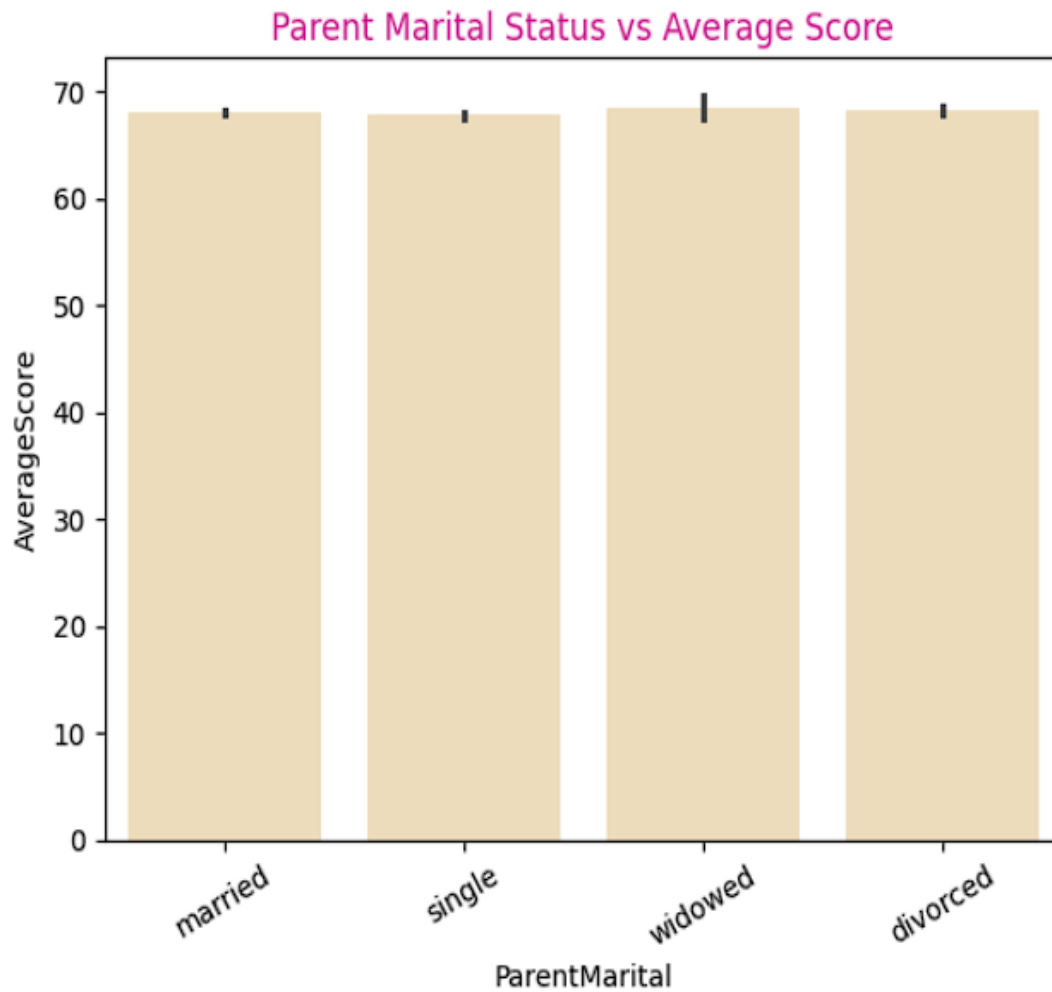
➤ Test Preparation

- Exploratory Data Analysis (EDA) was used to evaluate the impact of test preparation on student performance.
- Average scores were compared between students who completed test preparation and those who did not.
- Students who completed test preparation courses achieved higher average scores across subjects.
- The results indicate that structured test preparation has a positive impact on academic performance.



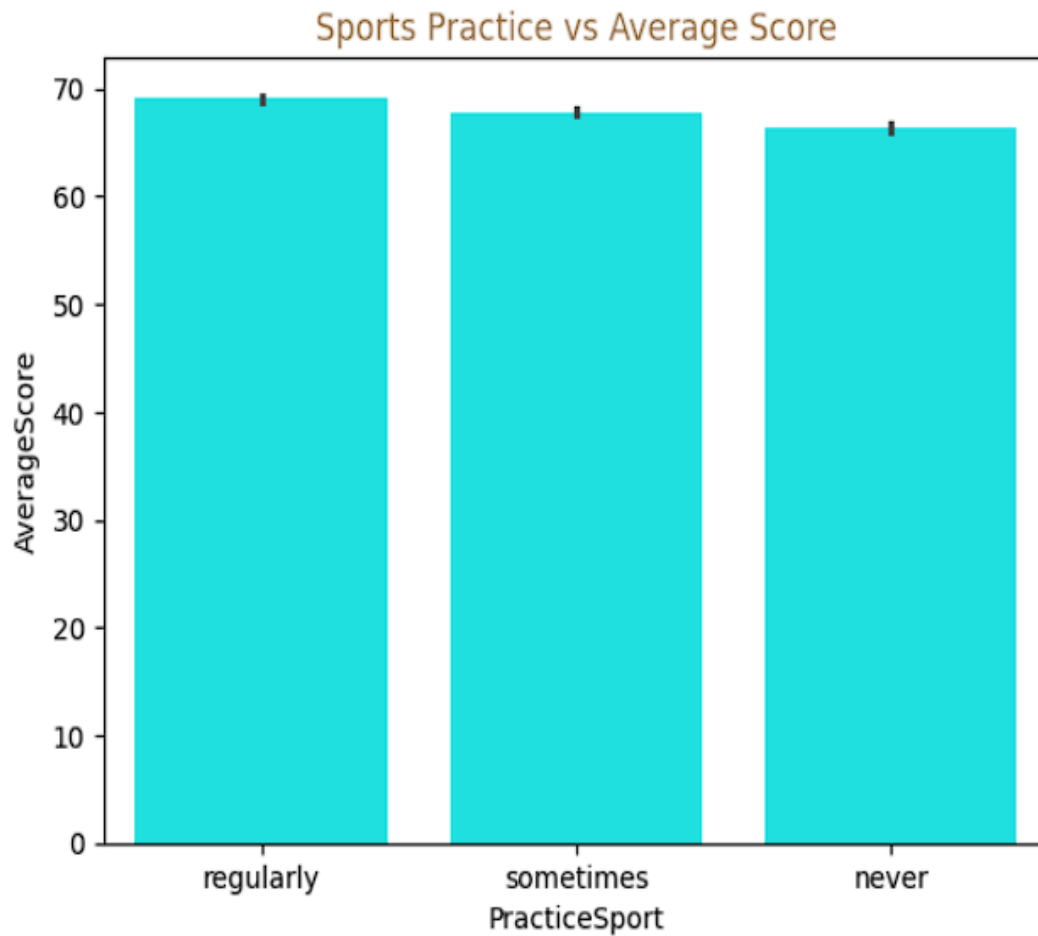
➤ Parent Marital Status

- EDA was conducted to examine the relationship between parent marital status and students' average academic scores.
- The analysis shows minimal variation in performance across different marital status categories, indicating a limited impact on overall student performance.



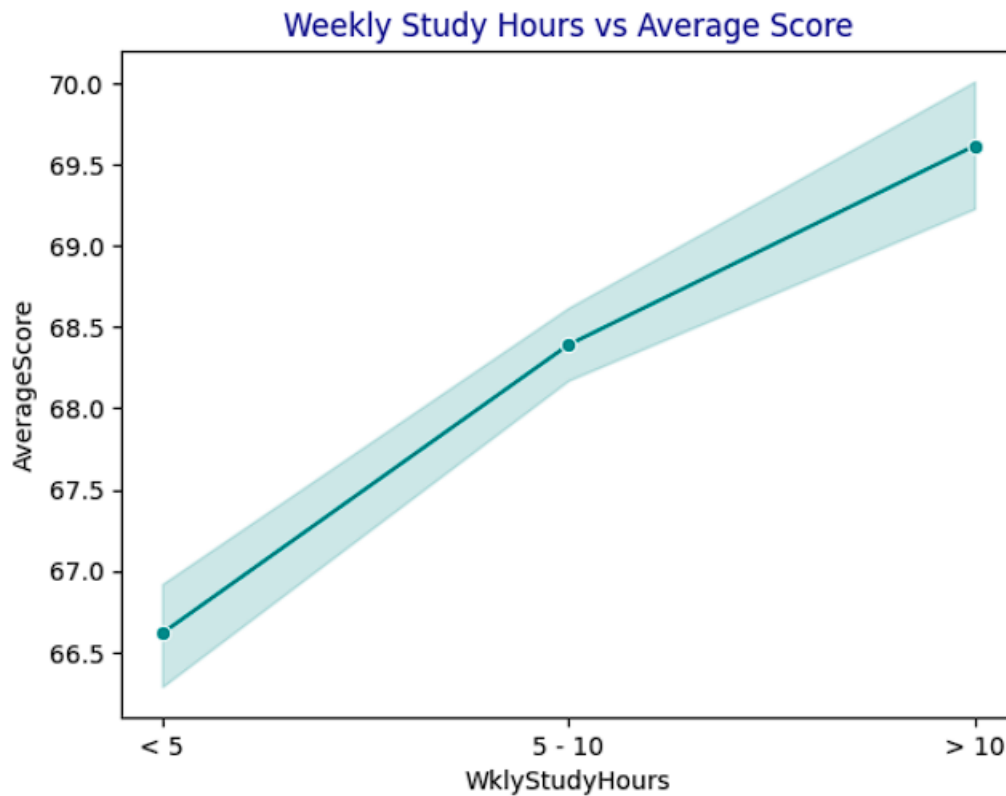
➤ Practice Sport

- EDA was used to analyse the relationship between sports practice frequency and students' average academic scores.
- Students who practice sports regularly show balanced and consistent academic performance compared to those who practice occasionally or not at all.



➤ Weekly Study Hours

- EDA was conducted to analyse the relationship between weekly study hours and students' average academic scores.
- The analysis shows a positive trend, where students who spend more time studying per week achieve higher academic scores.



8. Gender-wise Overall Performance Comparison

- Calculated average scores for male and female students.
- Visualized results using bar charts.
- Identified the gender with higher overall academic performance.

9. Key Insights & Findings

- Test preparation significantly improves student performance.
- Parental education plays an important role in academic success.
- Regular study hours positively impact scores.
- Female students generally scored slightly higher overall compared to male students.

- Sports participation supports balanced academic outcomes.

10. Conclusion

This project demonstrates that student performance is strongly influenced by gender, parental education, test preparation, study hours, and sports activity. The analysis shows that female students perform better overall, while structured test preparation and increased study time significantly enhance academic outcomes.